



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2015 Preliminary Course Assessment Task 1

Wednesday April 1, 2015

General instructions

- Working time – 70 minutes.
(plus 5 minutes reading time)
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 11M3A – Mr Hwang
- ☐ 11M3B – Mr Lin
- ☐ 11M3C – Mr Weiss
- ☐ 11M3D – Ms Ziazaris/Mr Weiss
- ☐ 11M3E – Mr Ireland
- ☐ 11M3F – Miss Lee

STUDENT #: (4xxxxxxxx) # SHEETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	6	7	Total	%
MARKS	$\overline{15}$	$\overline{16}$	$\overline{13}$	$\overline{10}$	$\overline{11}$	$\overline{12}$	$\overline{11}$	$\overline{88}$	

Question 1 (15 Marks)	Commence a NEW page.	Marks
(a) Consider the function $f(x) = \frac{2}{x^2 - 4}$		
i. Show that $f(x)$ is an even function		2
ii. What is the geometric significance of part (i)?		1
(b) State the domain and range of		
i. $y = -2x^2$		2
ii. $y = \sqrt{16 - x^2}$		2
iii. $y = -\sqrt{x + 2}$		2
(c) Sketch the graph of $y = 1 + \frac{1}{x + 3}$, showing the x -intercept, the y -intercept, and asymptotes.		3
(d) A baseball is projected vertically upwards and its height in metres after t seconds is given by $h = 30t - 5t^2$. Find the maximum height reached by the ball.		3

Question 2 (16 Marks)	Commence a NEW page.	Marks
(a) Sketch: $f(x) = \begin{cases} -1 & \text{for } x < -1 \\ 1 & \text{for } -1 \leq x \leq 1 \\ x^2 + 1 & \text{for } x > 1 \end{cases}$		3
(b) Sketch each of the following on separate number planes.		
i. $y = -2x^2$		3
ii. $xy = 2$		3
iii. $y = 6 - 3x $		2
(c) Given that, for all values of x , $f(x) = ax^2 + bx + c$ and that $f(0) = 6$, $f(1) = 12$ and $f(-1) = -4$, determine the values of a , b and c		3
(d) Find $f(x)$ if $f(x + 2) = 3x + 5$		2

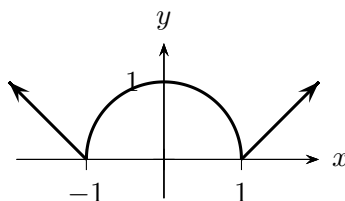
Question 3 (13 Marks)	Commence a NEW page.	Marks
(a) Find the value of a if $\sqrt{a} = 6\sqrt{3}$.		2
(b) Fully simplify: $\frac{a^2 - 2ab + b^2}{a^2 - b^2} - \frac{a^2 - b^2}{a^2 + 2ab + b^2}$		3
(c) Fully simplify: $\frac{3^{n+2} + 3^n}{3^n}$		2
(d) Solve for x :		
i. $x^4 = 4(x^2 + 3)$		3
ii. $3^{5x+7} = \frac{1}{81}$		3

Question 4 (10 Marks)

Commence a NEW page.

Marks

- (a) Sketch $y = 1 - 3^x$, showing all important features. **3**
- (b) The following diagram shows a sketch of $y = g(x)$.



Sketch, on separate diagrams (each being one third of a page):

- i. $y - 1 = g(x)$ **2**
- ii. $y = g(x - 1)$ **2**
- (c) Solve for x : $(x - 1)^2 (3 - x) > 0$ **3**

Question 5 (11 Marks)

Commence a NEW page.

Marks

- (a) Find the period and amplitude of the curve $y = 3 \sin(2x)$. **2**
- (b) i. Prove that $(\operatorname{cosec}^2 x - 1) \sin^2 x = \cos^2 x$. **2**
- ii. Hence or otherwise, solve **3**

$$(\operatorname{cosec}^2 x - 1) \sin^2 x = \frac{3}{4}$$

for $0^\circ \leq x \leq 360^\circ$.

- (c) A ship is sailing from on a bearing of 205° from port A to port B . Port B is 70 nm west of port A . **2**

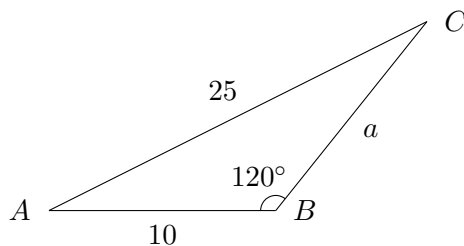
Find how far the ship has sailed.

- (d) If $\tan \theta = -\frac{4}{3}$ and $90^\circ \leq \theta \leq 180^\circ$, find the exact value of $\sin \theta$. **2**

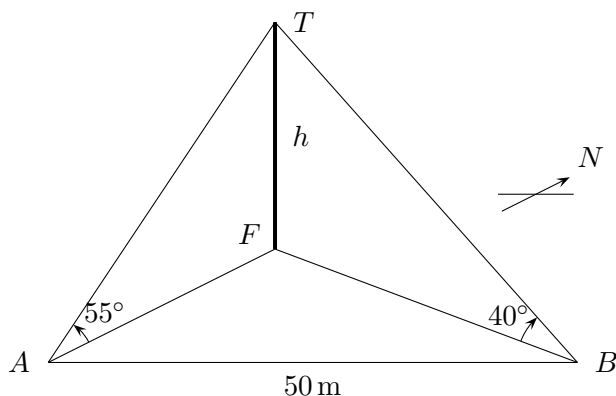
Examination continues overleaf...

- Question 6** (12 Marks) Commence a NEW page. **Marks**
- (a) i. Given that $f(x) = x^2 + 2$, find the largest possible domain consisting of positive real numbers such that $f(x)$ has an inverse function $f^{-1}(x)$. **2**
- ii. Find the equation of $f^{-1}(x)$. **2**
- iii. State the domain and range of both $f(x)$ and $f^{-1}(x)$. **2**
- iv. Sketch $f(x)$ and $f^{-1}(x)$ on the same set of axes, using the same scale for the x and y axes. **3**
- (b) Solve: $\frac{x^2 - 5x}{x - 4} \leq 3$. **3**

- Question 7** (11 Marks) Commence a NEW page. **Marks**
- (a) Sketch the graph of $y = 3 - 2 \cos 4x$ for $0^\circ \leq x \leq 180^\circ$, showing all important features. **3**
- (b) Find the *exact* value of a : **3**



- (c) The diagram shows a tower TF of height h metres standing due north of A on level ground. The angles of elevation of the top T of the tower from two points A and B (due east of A) on the ground are 55° and 40° respectively. The distance AB is 50 m.



- i. Find the size of $\angle FAB$. **1**
- ii. Find AF and BF in terms of h . **2**
- iii. Hence find the height of the tower to the nearest metre. **3**

End of paper.

Suggested Solutions

Suggested solutions

Question 8 (Weiss)

(a)

(a)

Question 10(Weiss)

Question 9 (Weiss)

(a)

Q1) a) i/ $f(x) = \frac{2}{x^2-4}$

$$f(-x) = \frac{2}{x^2-4}$$

$$= f(x)$$

∴ even function

ii/ symmetry about y-axis.

b) i/ $\{x: \text{all real } x\}$

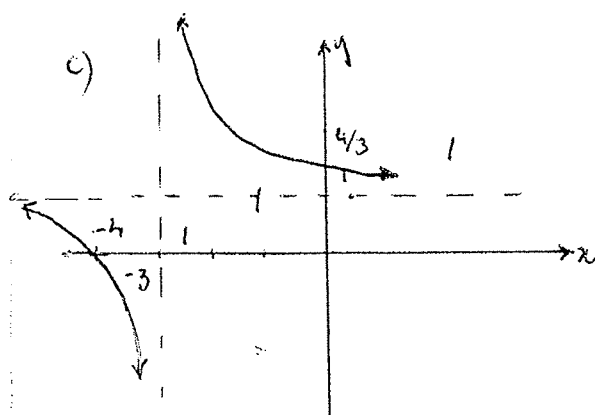
$$\{y: y \leq 0\}$$

ii/ $\{x: -4 \leq x \leq 4\}$

$$\{y: 0 \leq y \leq 4\}$$

iii/ $\{x: x \geq -2\}$

$$\{y: y \leq 0\}$$



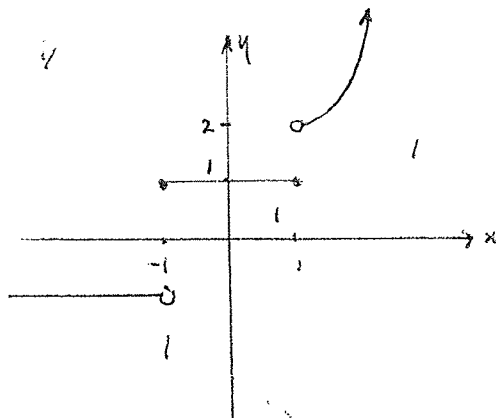
d) $h = 5t(6-t)$

when $t = 3$

$$h = 15(3)$$

$$= 45 \text{ m}$$

Q2) a) i



b)

$$f(x) = ax^2 + bx + c$$

$$f(0) = c = 6$$

$$f(1) = a + b + 6 = 12$$

$$a + b = 6 \quad (1)$$

$$f(-1) = a - b + 6 = -4$$

$$a - b = -10 \quad (2)$$

$$(1) + (2)$$

$$2a = -4$$

$$\therefore a = -2$$

$$b = 8$$

$$c = 6$$

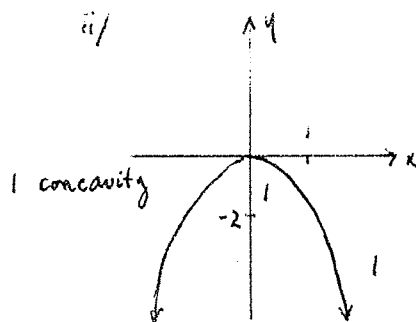
c)

$$f(x+2) = 3x+5$$

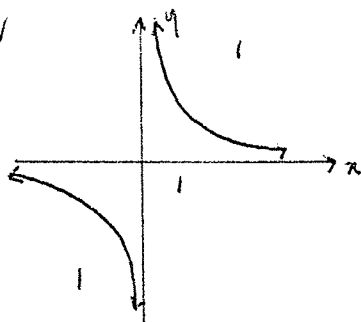
$$= 3(x+2) - 1$$

$$f(x) = 3x - 1$$

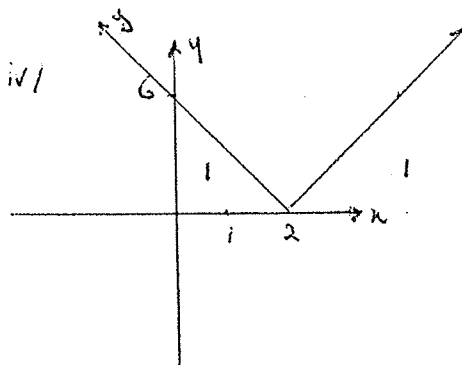
ii/



iii/



iv/



$$\begin{aligned} \text{Q3) a) } \sqrt{a} &= 6\sqrt{3} \\ &= \sqrt{36 \cdot 3} \quad | \\ \therefore a &= 108 \quad | \end{aligned}$$

$$\begin{aligned} \text{b) } & \frac{(a-b)^2}{(a+b)(a-b)} - \frac{(a-b)(a+b)}{(a+b)^2} \quad | \\ &= \frac{(a-b)^2(a+b) - (a-b)^2(a+b)}{(a+b)^2(a-b)} \quad | \\ &= 0 \quad | \end{aligned}$$

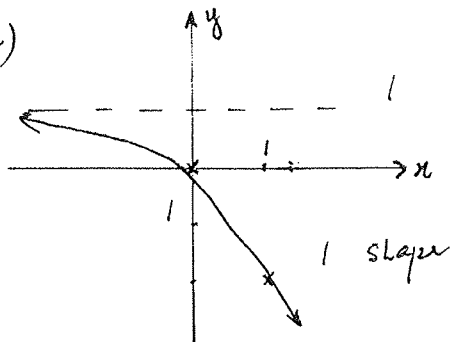
$$\begin{aligned} \text{c) } & \frac{3^n(3^3 + 1)}{3^n} \quad | \\ &= 10 \quad | \end{aligned}$$

$$\begin{aligned} \text{d) i/ } x^4 - 4x^2 - 12 &= 0 \\ (x^2 + 2)(x^2 - 6) &= 0 \quad | \\ \therefore x^2 &= 6 \text{ or } -2 \quad | \\ \therefore x &= \pm\sqrt{6} \text{ no real soln} \quad | \end{aligned}$$

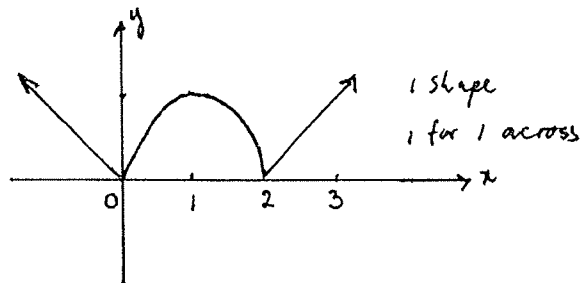
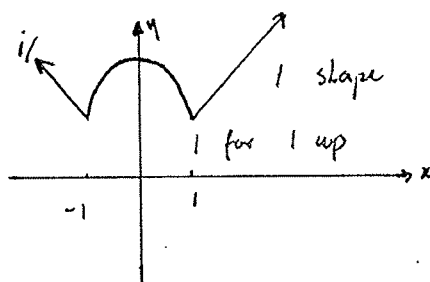
$$\begin{aligned} \text{ii/ } 3^{5x+7} &= 3^{-4} \quad | \\ 5x+7 &= -4 \\ 5x &= -11 \quad | \\ x &= \frac{-11}{5} \quad | \end{aligned}$$

Q4)

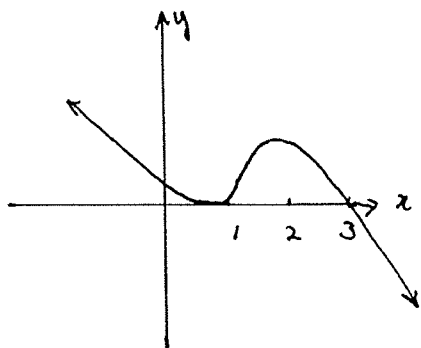
a)



b)



c)



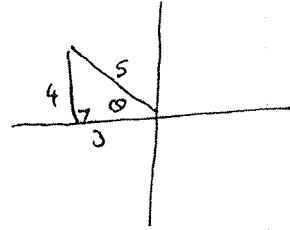
$$\therefore f(x) > 0$$

$$x \leq 3 \text{ but } x \neq 1$$

1

1

Q5) a) amplitude is 3 | d)
period 180° |



$$\sin \theta = \frac{4}{5}$$

+ve |

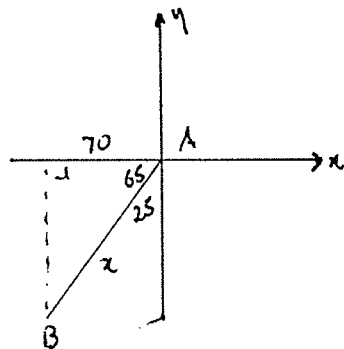
b) i/ L.H.S: $(\operatorname{cosec}^2 x - 1) \sin^2 x$ |
 $= \cot^2 x \cdot \sin^2 x$ |
 $= \cos^2 x$ |
 $= \text{RHS}$ |

ii/ $\cos^2 x = \frac{3}{4}$ |

$$\cos x = \pm \frac{\sqrt{3}}{2}$$

$$\therefore x = 30, 150, 210, 330^\circ$$

c)



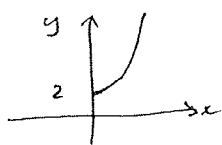
$$\frac{70}{x} = \cos 65^\circ$$

$$x = \frac{70}{\cos 65^\circ}$$

$$= 165.6 \text{ nm to 1D}$$

Q6

(a) (i) $f(x) = x^2 + 2$



$\therefore D: x \geq 0$

✓✓

(ii) For inverse, $x = y^2 + 2$

$\therefore y^2 = x - 2$

$y = +\sqrt{x-2} \text{ or } -\sqrt{x-2}$

} ✓

Since the domain of $f(x)$ is $x \geq 0$

$\therefore$ the range of $f^{-1}(x)$ is $y \geq 0$

$\therefore f^{-1}(x) = \sqrt{x-2}$

✓

(iii) For $f(x)$, $D: x \geq 0$

$R: y \geq 2$

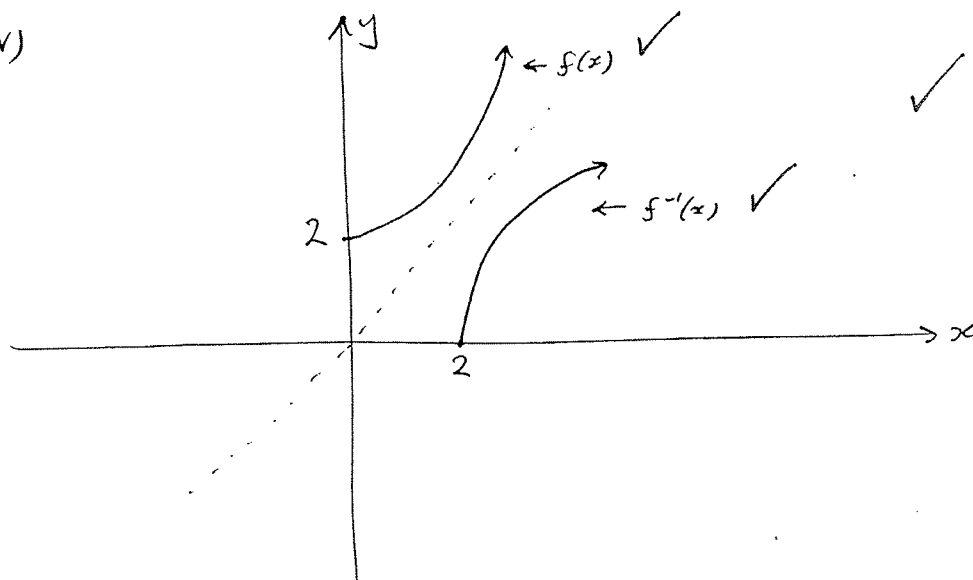
✓

For $f^{-1}(x)$, $D: x \geq 2$

$R: y \geq 0$

✓

(iv)



✓ needs:
 • symmetry
 • intercepts
 • shape.

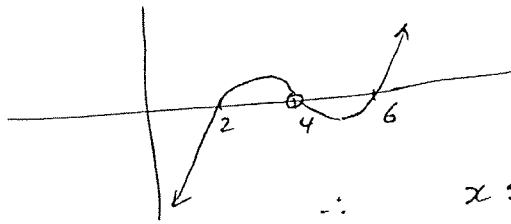
$$(b) \quad \frac{x^2 - 5x}{x - 4} \leq 3, \quad x \neq 4$$

$$(x^2 - 5x)(x - 4) \leq 3(x - 4)^2 \quad \checkmark$$

$$(x - 4)(x^2 - 5x - 3(x - 4)) \leq 0$$

$$(x - 4)(x^2 - 8x + 12) \leq 0$$

$$(x - 4)(x - 6)(x - 2) \leq 0 \quad \checkmark$$



$$\therefore x \leq 2 \text{ or } 4 < x \leq 6. \quad \checkmark$$

ALTERNATIVES

(1) Critical points: $x = 4$ and $\frac{x^2 - 5x}{x - 4} = 3$
 $\hookrightarrow x = 2, 6.$

Test several points:

eg $x = 0 \rightarrow \text{LHS} = \frac{0 - 0}{-4} = 0 \leq 3 \quad \therefore \checkmark$

$x = 3 \rightarrow \text{LHS} = \frac{9 - 15}{3 - 4} = 6 \not\leq 3 \quad \therefore \times$

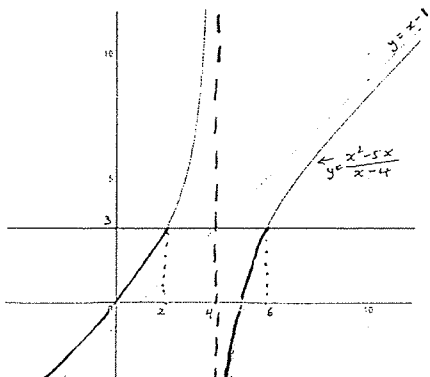
$x = 5 \rightarrow \text{LHS} = \frac{25 - 25}{1} = 0 \leq 3 \quad \therefore \checkmark$

$x = 10 \rightarrow \text{LHS} = \frac{50}{6} \not\leq 3 \quad \therefore \times$

Also, $x = 2$ & $x = 6$ give $\text{LHS} \leq 3.$

$$\therefore x \leq 2 \text{ or } 4 < x \leq 6. \quad \checkmark$$

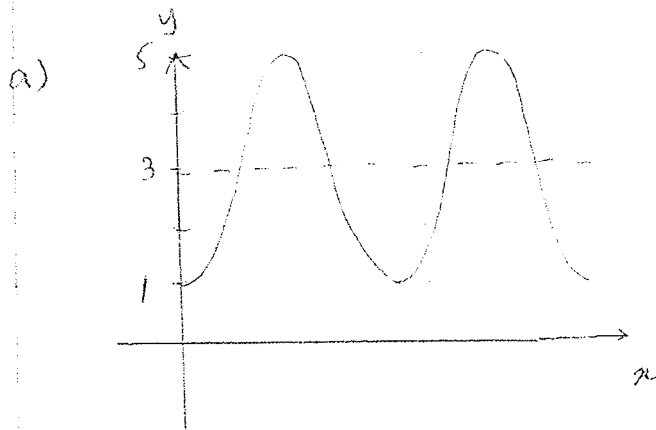
(2) Graph:



$\checkmark \checkmark$

$$\rightarrow \therefore x \leq 2 \text{ or } 4 < x \leq 6. \quad \checkmark$$

Question 7



b) From the cosine rule

$$25^2 = a^2 + 10^2 - 2(a)(10) \cos 120^\circ$$

$$25^2 - 10^2 = a^2 - 20a \cos 120^\circ$$

$$a^2 + 10a - 525 = 0$$

$$a = \frac{-10 \pm \sqrt{2200}}{2}$$

$$\text{as } a > 0, \quad a = 5\sqrt{22} - 5$$

c) i) $\angle FAB = 90^\circ$

$$\text{ii) } \tan 55^\circ = \frac{h}{AF}$$

$$\therefore AF = h \cot 55^\circ$$

$$\tan 40^\circ = \frac{h}{BF}$$

$$BF = h \cot 40^\circ$$

$$\text{iii) } 50^2 + (h \cot 55^\circ)^2 = (h \cot 40^\circ)^2$$

$$h^2 (\cot^2 40^\circ - \cot^2 55^\circ) = 50^2$$

$$h^2 = \frac{50^2}{\cot^2 40^\circ - \cot^2 55^\circ}$$

$$\therefore h \approx 52 \text{ m}, \quad h > 0$$